

Multivariable Control of a Nonlinear Industrial Evaporator

PI Control, Gain Scheduling, State Feedback and LQR

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Abstract

This report investigates the design and evaluation of control strategies for an industrial evaporator process. Decentralized PI controllers were first developed to regulate product composition, operating pressure, and separator level. Robustness studies revealed sensitivity to thermal parameter variations and actuator saturation, motivating the implementation of anti-windup protection.

To improve performance over a wider operating range, three advanced control strategies were evaluated: gain scheduling, multivariable state feedback, and Linear Quadratic Regulator (LQR) control. Their performance was assessed in terms of transient response, disturbance rejection, robustness to process uncertainty, and operation under varying production conditions.

The results demonstrate that gain scheduling significantly improves performance across different operating points, while multivariable state-feedback methods provide superior coordination between interacting process variables. Among the methods investigated, LQR achieves the best overall balance between tracking performance, robustness, and control effort. Consequently, LQR is a promising candidate for further development; integral augmentation and validation on the full constrained model remain necessary before industrial implementation.

1 Introduction

This report presents the development of a control system for an industrial evaporator. Following initial validation of the dynamic model and implementation of a basic level controller for the separator, the control system was extended to regulate two additional key variables — product composition (X_2) and operating pressure (P_2), with P_{100} and F_{200} as the corresponding actuators. All process variables and their nominal values used are given in Table 3 in Appendix.

The system's robustness was assessed by varying key process parameters such as density, mass, and heat transfer coefficient. Integrator wind-up was also addressed by applying output saturation limits to prevent instability and steady-state error.

Building on this foundation, three advanced control strategies were implemented: gain scheduling, multivariable pole placement, and Linear Quadratic Regulator (LQR) control with integral action. The report concludes with a comparative evaluation of these methods in terms of stability, responsiveness, robustness, and implementation complexity, to identify the most suitable approach for the system.

2 Baseline Decentralised PI Control

In addition to the controller for L_2 , in order to ensure better stability, we also implemented controllers for X_2 and P_2 . At the end of this section, we have three well-tuned controllers: L_2 (F_2 as control variable); X_2 (P_{100} as control variable); P_2 (F_{200} as control variable).

2.1 Controllers for P_2

We designed a PI controller for the operating pressure P_2 , using the cooling water flow rate F_{200} as the control variable. By linearising the pressure loop about the nominal operating point, we obtained the following parameters:

$$K_p = -785.2, \quad T_i = 12.1$$

2.2 Controller for X_2

We also designed, implemented and tested a PI controller for the product composition X_2 , using the steam pressure P_{100} as the control variable. We used the Ziegler-Nichols procedure, which gradually increases proportional gain until X_2 starts to oscillate with constant amplitude.

We found that the ultimate gain $K_u = 12.5$, and the period of oscillation is $T_u = 14.0$. Consequently, we set the proportional gain to $K_p = K_u/2.2 = 5.68$, and the integral time constant to $T_i = T_u/1.2 = 11.67$.

2.3 Testing stability

We tested the stability of the L_2 loop with X_2 and P_2 controllers in place. We linearised the model for the L_2 - F_2 open loop, and found that the phase margin is 17.3 degrees. We changed the proportional gain of the L_2 controller from -14.16 to -28 to raise the phase margin to 25 degrees.

As a result, we obtained 3 controller for L_2 , X_2 , P_2 , where

$$G_{L_2}(s) = -28 \left(1 + \frac{1}{18.96s} \right), \quad G_{P_2}(s) = -785.2 \left(1 + \frac{1}{12.10s} \right), \quad G_{x_2}(s) = 5.68 \left(1 + \frac{1}{11.67s} \right)$$

3 Robustness Investigation to Process Parameter Variations

A real industrial plant never behaves exactly as its mathematical idealization. To ensure that the decentralized PI controllers designed at the nominal operating point are reliable under model-plant mismatches, a single-parameter sensitivity analysis was performed. The three key process parameters were varied individually within their operating bounds used in the sensitivity study: $\rho_A \in [10, 40]$, $M \in [15, 25]$, and $UA_2 \in [5.0, 8.0]$, while keeping the remaining parameters at their nominal values ($\rho_{A,\text{nom}} = 20$, $M_{\text{nom}} = 20$, $UA_{2,\text{nom}} = 6.84$).

The closed-loop dynamic responses of the three controlled outputs—Product Composition X_2 , Operating Pressure P_2 , and Separator Liquid Level L_2 —were recorded over a 500-minute simulation horizon. The comprehensive evaluation is detailed below.

3.1 Case 1 & 2: Robustness to Material Parameters (ρ_A and M)

The system performance under variations of the material sector parameters—namely the component density ρ_A and the separator hold-up mass M —is illustrated in Figure 12 and Figure 16.

An inspection of the trajectories shows that the responses under varying ρ_A (ranging from 10 to 40) and M (ranging from 15 to 25) yield nearly identical and tightly superimposed curves. The change in equipment mass inventory and density has negligible influence on the transient path. For both parameter groups, the composition loop smoothly tracks back towards its state, and the liquid level L_2 remains well-regulated. This exceptional overlap indicates that the decentralized control loops exhibit excellent robust performance and high stability margins against mass-balance variations.

3.2 Case 3: Sensitivity to Thermal Parameter (UA_2) and Actuator Saturation

In stark contrast to the material parameters, the system exhibits high sensitivity to variations in the condenser heat transfer coefficient UA_2 , as depicted in Figure 4.

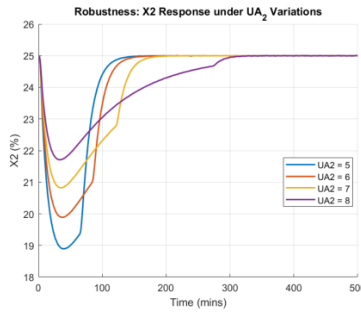


Figure 1: X_2 response (UA_2 variation).

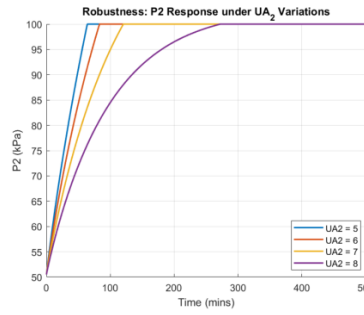


Figure 2: P_2 response (UA_2 variation).

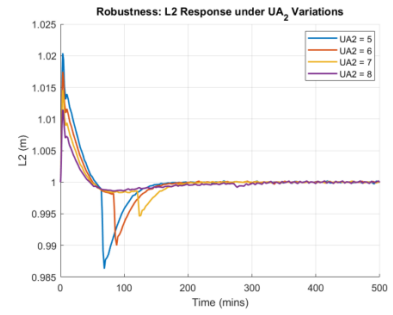


Figure 3: L_2 response (UA_2 variation).

Figure 4: CClosed-loop responses under variations of condenser thermal parameter UA_2 .

The transient responses demonstrate a clear trade-off between the settling time and the dynamic deviation as UA_2 varies. Crucially, across all scenarios, the operating pressure P_2 does not return to its original nominal value but instead drives into a hard saturation constraint at exactly 100 kPa, reflecting the physical limitation of the cooling water actuator.

- **Increase in UA_2 ($UA_2 = 8$):** As UA_2 increases, the system exhibits a much longer settling time. The time required for the controlled variables (particularly composition X_2) to stabilize increases significantly. However, this slower transition effectively smooths out aggressive deviations, resulting in a substantially smaller undershoot.
- **Decrease in UA_2 ($UA_2 = 5, 6$):** Conversely, as UA_2 decreases, the speed of response accelerates, forcing the system to reach its target limit faster. However, this comes at the cost of transient control quality, leading to a much larger undershoot. For instance, at $UA_2 = 5$, X_2 plunges deeply to 18.9%, and a severe cross-coupling disturbance is propagated into the liquid level L_2 loop, inducing a sudden dip to 0.986 m.

3.3 Robustness Synthesis and Conclusion

The single-parameter investigation demonstrates an asymmetric robust performance profile. While the control framework is highly insensitive to mass-balance shifts (ρ_A, M), it displays a compromised transient performance margin against energy-balance perturbations (UA_2).

Furthermore, the persistent flat-lining of P_2 at 100 kPa under all variations uncovers a severe susceptibility to actuator saturation. This persistent discrepancy from the nominal steady-state strongly suggests that the P_2 loop is highly vulnerable to *integrator wind-up*. The accumulation of error during this saturated period explains the extended settling times, highlighting the critical necessity of implementing an anti-windup protection scheme to safeguard the multivariable process performance.

4 Actuator Saturation and Integrator Wind-up

A severe vulnerability in decentralized PI control is integrator wind-up. When a large disturbance drives the controller output into its physical actuator limits, the integral term continues to accumulate tracking error blindly. This results in an excessively inflated integral state, causing drastic overcorrection, severe transient lag, or complete loop destabilization once the disturbance is removed.

To evaluate this behavior, the closed-loop system was subjected to a regulatory test: a pulse disturbance raising feed composition X_1 from 5% to 25% (a 20 percentage-point increase) lasting for 100 minutes. The transient responses of the product composition X_2 over a 1000-minute horizon are compared in Figure 5.

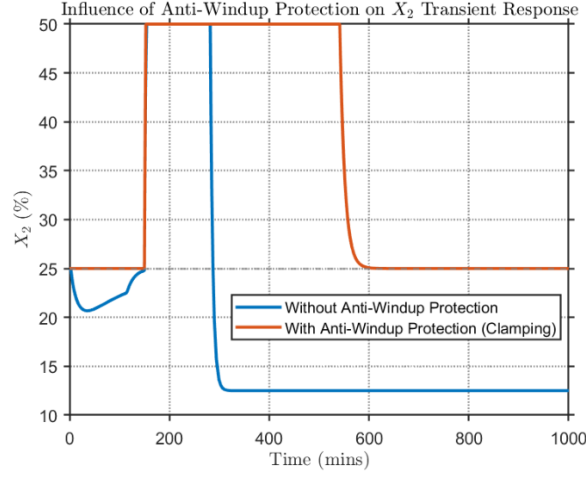


Figure 5: Comparison of X_2 loop performance under a 100-minute X_1 pulse disturbance with and without clamping anti-windup protection.

The simulation trajectories demonstrate two distinct behaviors:

- **Without Anti-Windup Protection (Blue Curve):** Due to the unconstrained error accumulation during the disturbance phase, the saturated integrator severely overcorrects after the pulse ends. Coupled with strong multi-variable process interactions, the system fails to recover and instead tracks to an incorrect, non-nominal steady state ($X_{2,ss} \approx 12.5\% \neq 25\%$), compromising the entire network.
- **With Anti-Windup Protection (Orange Curve):** To resolve this, physical saturation bounds were applied to the controllers ($L_2 \in [0, 1.9]$, $X_2 \in [0, 45]$, $P_2 \in [0, 90]$) along with a *clamping (conditional integration)* algorithm. The moment an actuator saturates, further integration is frozen. Consequently, the controller exits saturation instantly when conditions normalize, allowing X_2 to smoothly back away from its ceiling and converge back to the nominal 25% setpoint.

The improved recovery validated in Figure 5 indicates that constraint management is crucial for this multivariable process. Therefore, the clamping anti-windup architecture is fully adopted for all subsequent stages of this project.

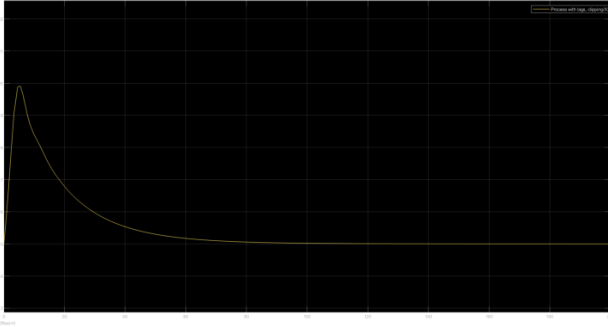
5 Performance under Varying Operating Points

To verify the system robustness across different operating points, we adjusted the setpoint of X_2 (nominal value: 25%) and observed the responses of L_2 , X_2 and P_2 .

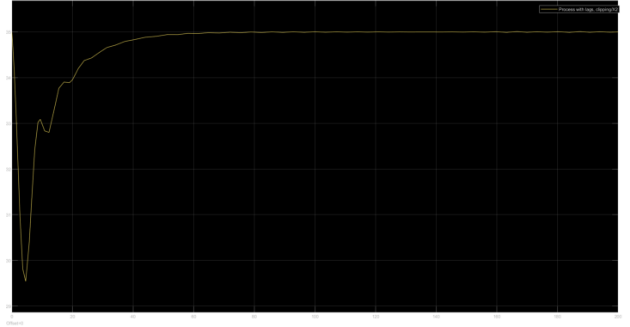
When X_2 setpoint is 15%, all three variables converge to their nominal values, while X_2 has an overshoot of 4%. When the setpoint decreases further to 12.5%, the system fails to track the target.

A similar trend is observed when increasing X_2 . At $X_2 = 35\%$, the signal has an undershoot of 5% but converges correctly. The system loses tracking ability when X_2 exceeds 52%. The above phenomena are illustrated in Figures 3a and 3b.

P_2 produces a small steady-state error when X_2 deviates from 25%, which is less obvious than that of X_2 . L_2 always converges to its nominal value 1, while the settling time increases as the operating point moves away from 25%.



(a) Localized transient behavior at $X_2 = 15\%$



(b) Localized transient behavior at $X_2 = 35\%$

Figure 6: Dynamic regulatory profiles of the evaporator asset operating at shifted steady-state coordinates, demonstrating the severe non-linear degradation of the fixed nominal PI network.

The nominal value of P_2 is 50.5. The system cannot converge when P_2 exceeds 95. For L_2 , it only works normally within the range $0 \sim 1.9$; values higher than 1.9 are clipped by saturation limits at 2.

6 Gain-Scheduled Controller Design

6.1 Design Strategy and Simulink Implementation

To counteract the wide-envelope performance degradation and coupling identified in Section 5, a 1-D Gain Scheduling (GS) architecture was implemented on the X_2 loop. Using the product composition setpoint as the scheduling variable, 1-D Lookup Tables execute linear interpolation to dynamically adjust controller parameters across intermediate operating states. The scheduled nodes are anchored at the operational limits ($X_2 = 15\%$ and 35%) and the nominal baseline ($X_2 = 25\%$).

6.2 Parameter Synthesis: Linearization vs. Heuristic Tuning

An initial analytical iteration utilized MATLAB's `linmod` to extract open-loop state-space models at each node with adjacent loops closed. Theoretical PI parameters were synthesized targeting a uniform phase margin ($PM = 40.0^\circ$) and crossover frequency ($\omega_{cp} = 0.2497$ rad/min), as summarized in Table 1.

Table 1: Open-loop linearization results and theoretical PI parameters.

$X_{2,\text{set}}$ (%)	$F_{2,\text{ss}}$	$P_{100,\text{ss}}$	$F_{200,\text{ss}}$	DC Gain	K_{p_X2}	T_{i_X2} (min)	K_{i_X2}
15	3.3333	138.1000	101.4000	0.4586	6.5348	5.0044	1.3058
25	2.0000	194.7000	208.0000	0.4586	3.5171	3.6116	0.9738
35	1.4286	231.4000	320.4000	0.4586	2.2724	2.7015	0.8412

Full non-linear simulations revealed that these analytical gains were excessively aggressive, inducing severe actuator saturation ($P_{100} \in [0, 400]$) and loop instability. This failure highlights that localized linear metrics significantly underestimate multi-loop cross-talk and constraint boundaries.

To restore stability, an empirical tuning strategy was deployed to counter process non-linearities (sluggish tracking at low concentrations; over-excitation at high concentrations). Incorporating clamping anti-windup, a robust manually tuned gain schedule was established:

$$K_{p,\text{sched}} = [6.5, 5.318, 4.5], \quad T_{i,\text{sched}} = [11.17, 11.17, 11.17] \text{ min}$$

6.3 Final Results and Performance Analysis

The manual GS framework was validated against the fixed nominal PI baseline via symmetric setpoint migrations, with multivariable trajectories compiled in Figures 7 and 8.

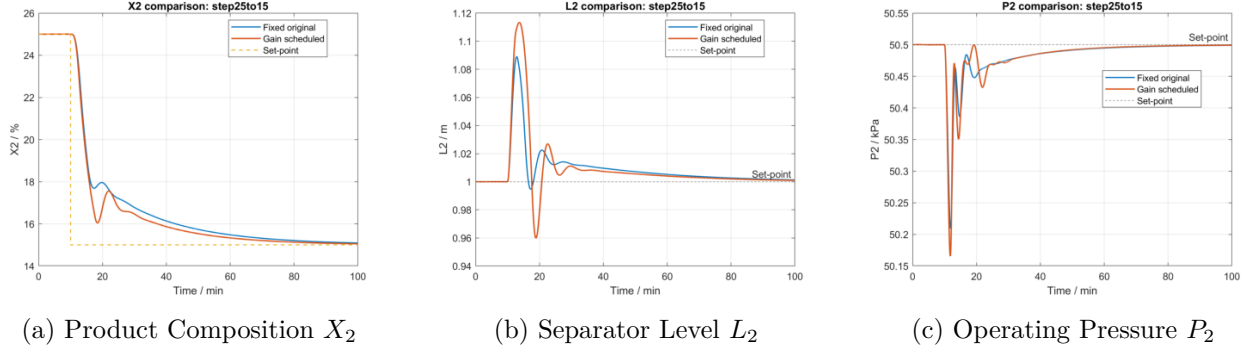


Figure 7: Process dynamic trajectories for a downward step transition (25% $\rightarrow$ 15%).

During the downward step (25% $\rightarrow$ 15%, Figure 7), the baseline fixed PI (blue) suffers from sluggish non-linear creep, requiring over 400 min to settle due to attenuated plant sensitivity. By scheduling an elevated gain ($K_p = 6.5$), the GS controller (orange) accelerates recovery, compressing the settling time to 150 min ($2.6\times$ speed enhancement). Although this rapid energy injection induces a brief transient spike in pressure ($\Delta P_2 \approx -1.3$ kPa), the disturbance is extinguished within 100 min, mitigating the prolonged cross-talk drift seen in the baseline.

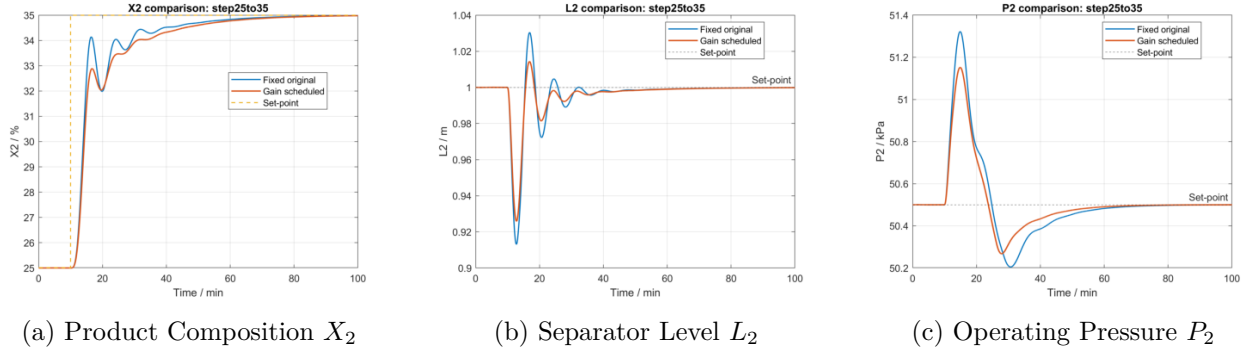


Figure 8: Process dynamic trajectories for an upward step transition (25% $\rightarrow$ 35%).

Conversely, during the upward step (25% $\rightarrow$ 35%, Figure 8), the increased high-concentration process gain causes the fixed PI to generate an excessive 16% overshoot relative to the 10 percentage-point step (peaking at 36.6%) and under-damped oscillations. Throttling the local gain down to $K_p = 4.5$ via the GS lookup tables enforces a smooth, near-monotonous convergence that restricts maximum overshoot below 4% of the step size (peaking at 35.4%).

Crucially, suppressing the tracking overshoot decouples transient shocks to adjacent loops; the massive pressure drop observed under fixed PI control ($\Delta P_2 \approx -5.0$ kPa) is curtailed to less than -2.5 kPa (a 50% disturbance reduction), while oscillatory cycling within the level loop L_2 is entirely eliminated.

In conclusion, quantitative empirical analysis confirms that the manual GS framework successfully delivers robust tracking acceleration, tight constraint containment, and effective cross-loop decoupling across the wide operational envelope.

7 State Feedback Control

7.1 Basic State Feedback

Multivariable state feedback control was subsequently implemented, defining the control input vector as $u = [F_2, P_{100}, F_{200}]^T$ and the state vector as $x = [X_2, P_2, L_2]^T$. The state feedback gain matrix, K , was designed via pole placement using the MATLAB `place` command. Using a deviation-variable

state-space representation, the basic state feedback controller takes the linear form:

$$\tilde{u} = -K\tilde{x}$$

Consequently, the closed-loop system dynamics are governed by:

$$\dot{\tilde{x}} = (A - BK)\tilde{x}$$

where the A and B matrices are derived from the reduced linearized plant model, K represents the pole-placement gain matrix, and $\tilde{u}$ and $\tilde{x}$ denote the input and state deviation vectors relative to their nominal steady-state values.

The pole-placement synthesis was conducted by sweeping candidate target locations, scaling from $\mathcal{O}(1/10)$ up to $\mathcal{O}(1)$. This sweeping strategy aims to achieve a stable yet sufficiently aggressive transient response to outperform the decentralized baseline in settling time, while preventing excessive under-damped oscillatory behavior. The relationship between pole locations in the complex plane and the corresponding damping factor is defined by:

$$s = \sigma \pm j\omega_d$$

$$\zeta = -\frac{\sigma}{\sqrt{\sigma^2 + \omega_d^2}}$$

Crucially, it must be recognized that the nominal linearization and subsequent pole placement were performed on an idealized open-loop model neglecting actuator dynamics. This abstraction allowed the full state vector to be reduced from $x = [X_2, P_2, L_2, F_{2,\text{lag}}, P_{100,\text{lag}}, F_{200,\text{lag}}]^T$ to the simplified vector $x = [X_2, P_2, L_2]^T$. This mismatch introduces a structural discordance between the idealized model and the true physical system containing unobserved actuator states. As a result, evaluating the controller on the full-lag plant yields three additional closed-loop poles alongside slightly shifted dominant poles. Let p denote the target poles utilized to compute K , and p_{lags} represent the actual closed-loop poles observed under full actuator lags:

$$p = \begin{bmatrix} -0.0900 & -0.0600 & -0.0300 \end{bmatrix},$$

$$p_{\text{lags}} = \begin{bmatrix} -0.0888 & -0.0603 & -0.0312 & -0.8022 & -0.8445 & -0.8288 \end{bmatrix}.$$

Notably, the three parasitic poles cluster near -0.8 , aligning closely with the reciprocal of the first-order actuator lag time constant ($T_i = 1.2$ min). This close match validates the theoretical assumption of unmodeled lag dynamics. A promising scope for a project extension involves designing a state observer to estimate these unobserved actuator states, thereby augmenting the feedback control law with a full-order state estimate.

The closed-loop transient trajectories resulting from the pole sweep are illustrated in Figure 20. Coincident poles at $p = [-0.43, -0.43, -0.43]$ were selected as the final design configuration due to their well-damped profile and well-bounded overshoot. Although the response velocity of this basic state feedback framework was satisfactory, Figure 20 reveals that the tracking trajectories fail to converge to their exact target values. This persistent steady-state offset establishes the primary motivation for introducing integral action.

8 Integral State Feedback

To eliminate the steady-state tracking errors characteristic of purely proportional state feedback, the control architecture was augmented by incorporating the integrals of the state tracking errors as auxiliary states. Defining v as the integral state vector satisfying $\dot{v} = \tilde{x}$, the augmented open-loop state-space representation is formulated as:

$$\begin{bmatrix} \dot{\tilde{x}} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} A & 0 \\ I & 0 \end{bmatrix} \begin{bmatrix} \tilde{x} \\ v \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} \tilde{u}$$

The generalized multivariable control law incorporating both state and integral feedback is expressed as:

$$\tilde{u} = - \begin{bmatrix} K & L \end{bmatrix} \begin{bmatrix} \tilde{x} \\ v \end{bmatrix}$$

This structure was implemented in Simulink by expanding the feedback path via a multiplexer (**Mux**) block, as illustrated in Figure 19.

The resulting dynamic responses, generated by assigning the augmented closed-loop poles to:

$$p = \begin{bmatrix} -0.1 & -0.2 & -0.3 & -0.4 & -0.5 & -0.6 \end{bmatrix},$$

demonstrate a substantial performance enhancement. Figures 21 and 22 compare the proportional state feedback (**sfprop**) against the augmented integral state feedback (**sfint**). The integral action successfully eliminates all steady-state tracking offsets across all setpoint step changes. More importantly, as demonstrated in the cross-coupling evaluation (Figure 22), the persistent deviations induced in L_2 and P_2 during X_2 setpoint migrations are reduced in the simulated steady state, marking a critical advancement in multi-loop decoupling.

However, while the integral state feedback successfully ensures zero steady-state error, it inherently increases the transient overshoot observed in the L_2 and P_2 channels during composition adjustments. This performance trade-off is highly visible in the P_2 response in Figure 22, where the heightened aggressiveness of the integrated error feedback induces more oscillatory behavior. These compromised dynamics motivate transitioning to Linear Quadratic Regulator (LQR) control, which offers a systematic framework for balancing tracking performance against control effort, eliminating the need for arbitrary pole trial-and-error.

9 Linear Quadratic Regulation

To build upon the state feedback paradigm, an optimal control framework was investigated. This methodology computes the state feedback gain matrix by minimizing a quadratic cost function J , wherein the state deviation errors and control input trajectories are explicitly penalized via the weighting matrices Q and R , respectively. The optimization problem is solved using the MATLAB `lqr()` function:

$$J = \int_0^\infty \left(\tilde{x}(t)^T Q \tilde{x}(t) + \tilde{u}(t)^T R \tilde{u}(t) \right) dt$$

Bryson's rule was initially adopted to synthesize the baseline Q and R matrices, assigning diagonal weights based on the inverses of the maximum acceptable squared state deviations and control inputs:

$$Q = \text{diag} \left(\frac{1}{x_{i,\max}^2} \right), \quad R = \text{diag} \left(\frac{1}{u_{j,\max}^2} \right)$$

Although this baseline design provided an effective initial synthesis, the resulting closed-loop responses exhibited several undesirable transient properties (as shown by the cyan trajectory in Figure 23), necessitating empirical tuning.

Specifically, the baseline LQR formulation placed insufficient penalty on tracking errors, resulting in poor convergence characteristics in the X_2 and P_2 step responses. To prioritize error attenuation, the Q matrix was incrementally scaled by factors of 10 until it approached the marginal stability boundary of the plant. While this adjustment yielded significantly improved tracking rates, the lingering performance differences among individual step responses remained sub-optimal. Further specialized tuning iterations yielded diminishing returns, resulting in the selection of the final Q and R

configuration:

$$Q_{\text{final}} = 100 Q_{\text{base}} = 100 \begin{bmatrix} \frac{1}{25^2} & 0 & 0 \\ 0 & \frac{1}{50.5^2} & 0 \\ 0 & 0 & \frac{1}{1^2} \end{bmatrix}, \quad R_{\text{final}} = R_{\text{base}} = \begin{bmatrix} \frac{1}{2^2} & 0 & 0 \\ 0 & \frac{1}{190^2} & 0 \\ 0 & 0 & \frac{1}{200^2} \end{bmatrix}.$$

10 Comparative Evaluation

10.1 Transient Response

The decentralized PI controller provided satisfactory regulation near the nominal operating point but exhibited slow responses and degraded performance away from the design condition.

Gain scheduling improved performance across the operating range by adapting controller parameters to the desired composition. Settling time was significantly reduced and overshoot was suppressed at higher operating points.

Basic state feedback achieved faster responses through multivariable coordination, but steady-state offsets remained. The addition of integral action removed these offsets and improved setpoint tracking, although at the expense of slightly larger overshoots.

Among the methods investigated, LQR delivered the best overall transient performance, providing fast, well-damped responses while maintaining reasonable control effort.

10.2 Robustness to Operating Point Changes

The fixed-gain PI controller was only effective near the nominal operating point. Performance deteriorated as the operating condition moved away from $X_2 = 25\%$.

Gain scheduling provided the strongest robustness to operating-point variation, maintaining satisfactory performance across the required range of $X_2 = 15\% - 35\%$. State-feedback and LQR controllers remained stable over moderate operating-point changes but were inherently limited by their single-point linearisation.

10.3 Robustness to Parameter Variations

The baseline PI tests showed limited sensitivity to variations in ρ_A and M . The dominant source of performance degradation was variation in the heat-transfer coefficient UA_2 , which increased loop interaction and actuator saturation.

With anti-windup protection enabled, the disturbance-recovery test improved relative to the unprotected PI baseline. A common parameter sweep for every advanced controller would be needed to establish their relative robustness.

10.4 Implementation Complexity

Table 2 summarises the relative strengths of the investigated control strategies.

Table 2: Comparison of advanced control strategies

Method	Performance	Robustness	Complexity
Gain Scheduling	Moderate	High	Low
State Feedback	High	Moderate	Moderate
State Feedback + Integral	High	Good	Moderate
LQR	Highest	Good	High

Gain scheduling is simple to implement and performs well over a wide operating range. State-feedback methods offer improved dynamic performance but require an accurate plant model. LQR requires the greatest design effort but provides the most systematic and effective overall control solution.

10.5 Recommended Industrial Implementation

For applications prioritising simplicity and wide operating-range performance, gain scheduling is an attractive option. However, when maximum control performance is required and a reliable process model is available, LQR is preferred.

The reported LQR formulation uses the three process states. Combining it with integral augmentation is a proposed extension, rather than a fully documented result of this study. Industrial implementation would additionally require validation of actuator constraints and model uncertainty.

11 Conclusion

A complete control system has been developed for the industrial evaporator process, progressing from decentralized PI control to advanced multivariable control strategies.

Initial investigations demonstrated that the nominal PI controllers provide satisfactory regulation near the design operating point but suffer performance degradation under large operating-point changes, process uncertainty, and actuator saturation. The introduction of clamping anti-windup protection significantly improved disturbance recovery and ensured reliable operation under actuator constraints.

Gain scheduling successfully extended the operating range of the decentralized control structure by adapting controller parameters to different product specifications. State feedback control improved dynamic performance through coordinated multivariable action, while integral augmentation eliminated steady-state tracking errors. Finally, LQR provided a systematic optimisation framework that balanced state regulation and control effort, yielding the most favourable overall transient behaviour.

The comparison supports further investigation of LQR for this process. The available simulation results do not establish a general robustness guarantee or readiness for industrial deployment.

Future work could include observer design for unmeasured states, gain-scheduled LQR architectures, and more advanced anti-windup schemes to further improve performance under severe operating constraints.

A Closed-loop responses under variations of ρ_A and M

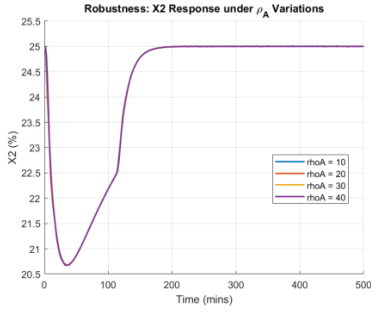


Figure 9: X_2 response (ρ_A variation).

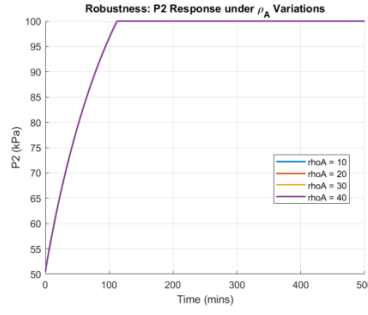


Figure 10: P_2 response (ρ_A variation).

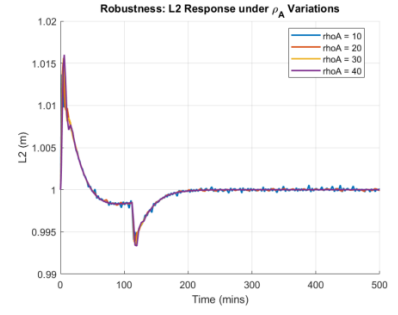


Figure 11: L_2 response (ρ_A variation).

Figure 12: CClosed-loop responses under variations of material density ρ_A .

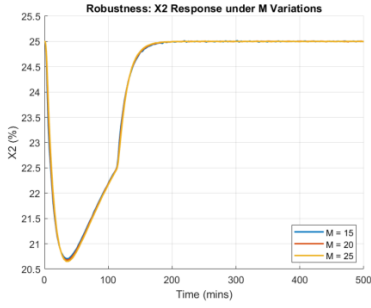


Figure 13: X_2 response (M variation).

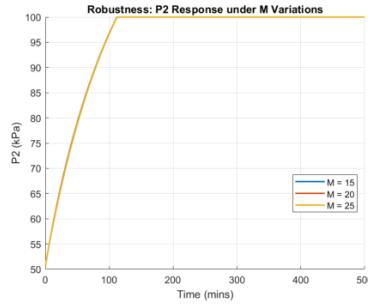


Figure 14: P_2 response (M variation).

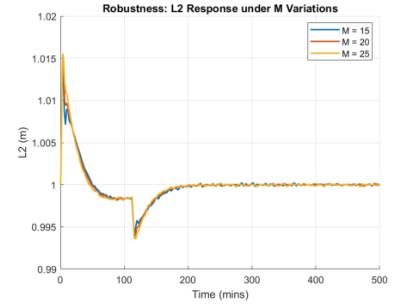


Figure 15: L_2 response (M variation).

Figure 16: CClosed-loop responses under variations of liquid hold-up mass M .

B Schematics

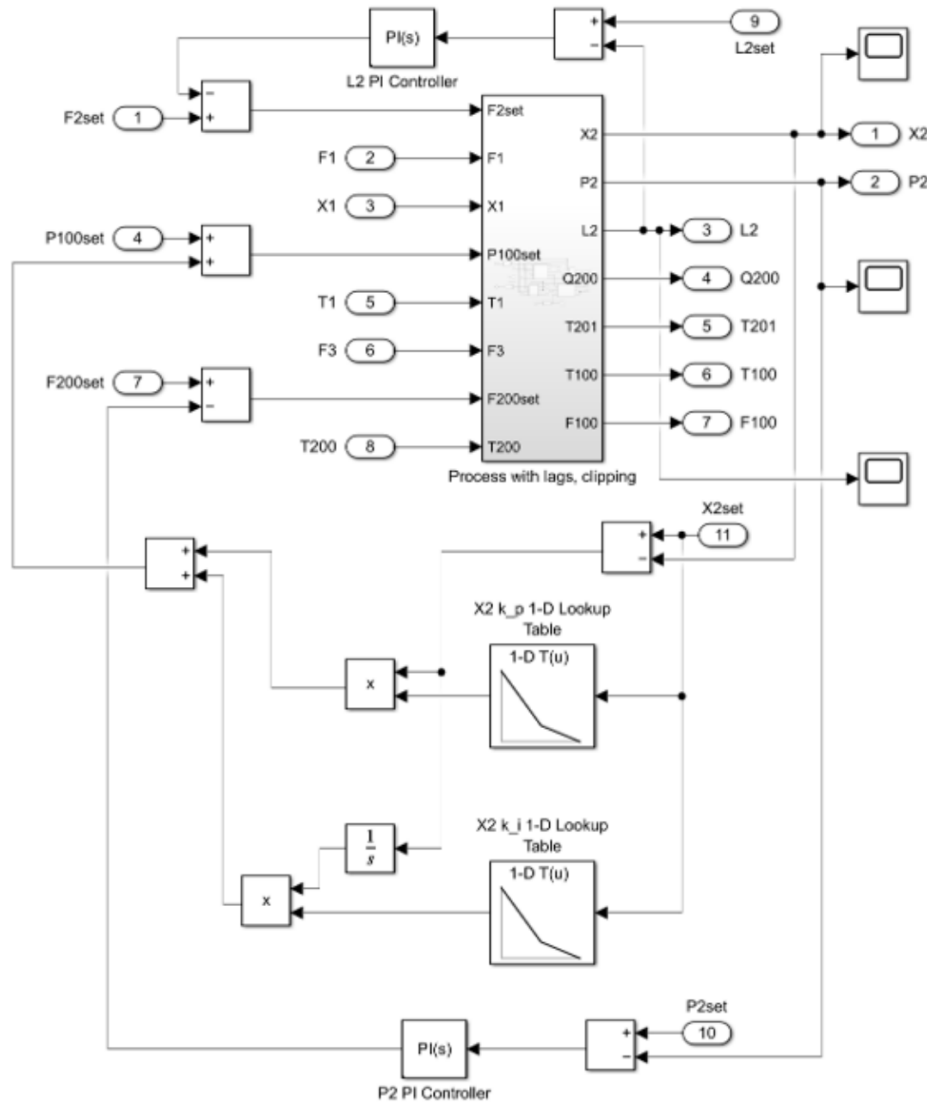


Figure 17: Simulink implementation of the 1-D lookup table framework for the X_2 gain-scheduled controller.

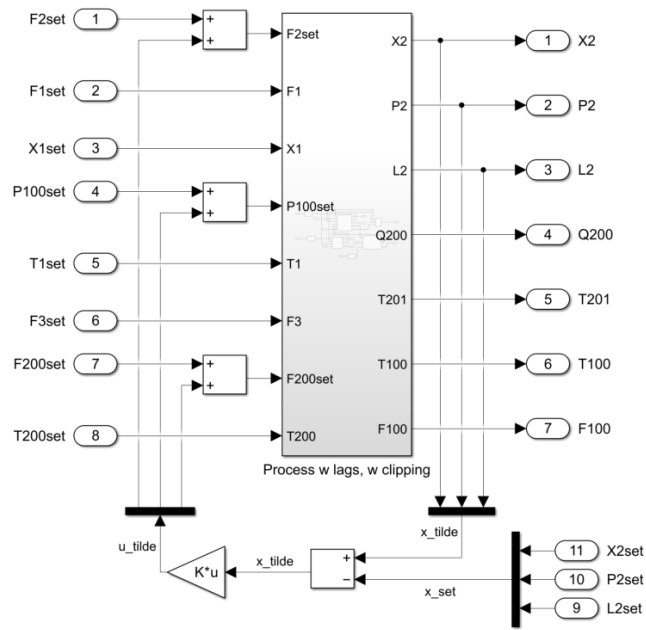


Figure 18: Basic State Feedback

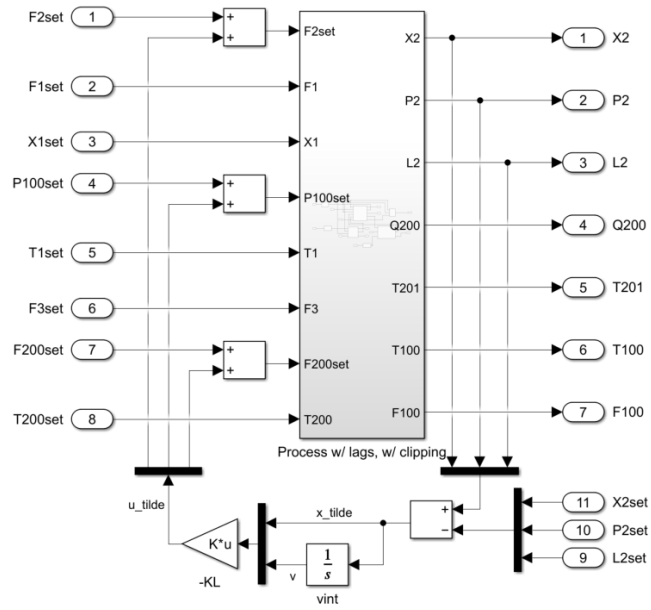
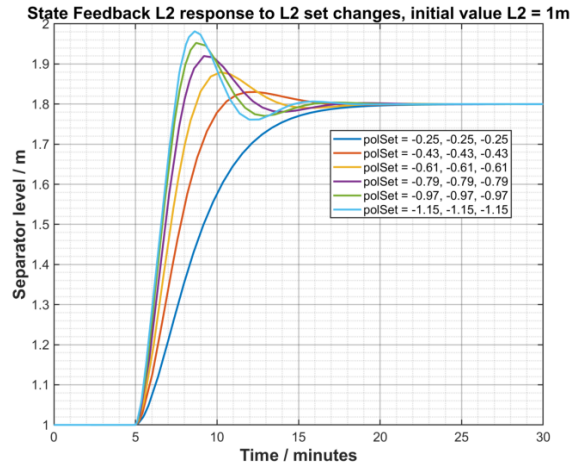


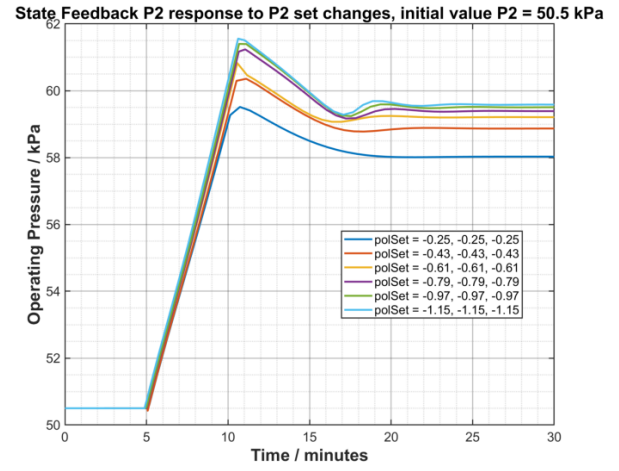
Figure 19: State Feedback With Integral Control

C State Feedback

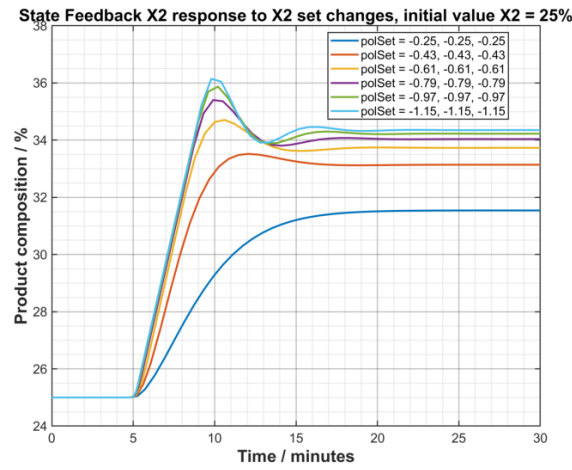
C.1 Poles sweep



(a) L2



(b) P2



(c) X2

Figure 20: X2 step down responses

C.2 State Feedback with Integral Action

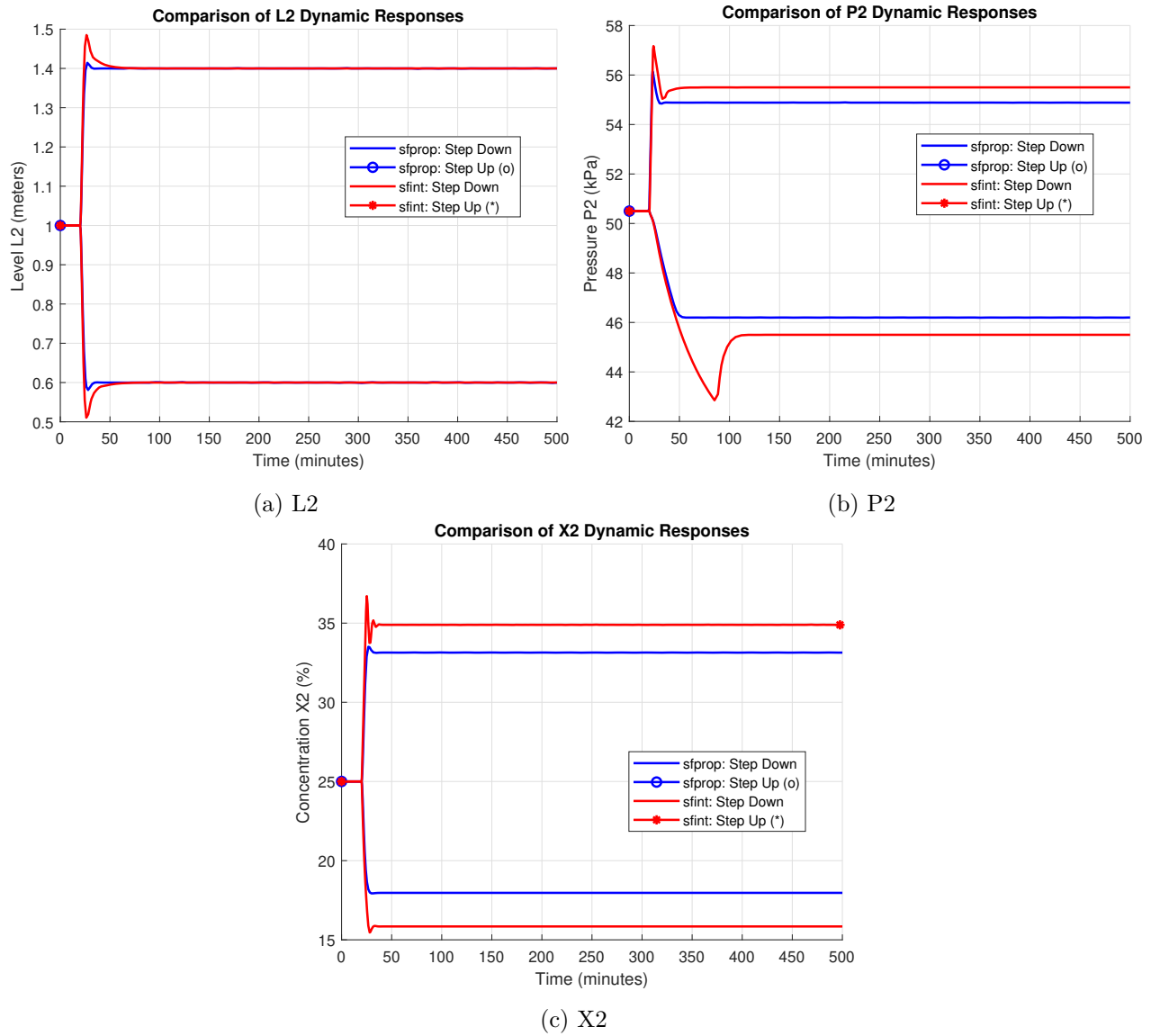
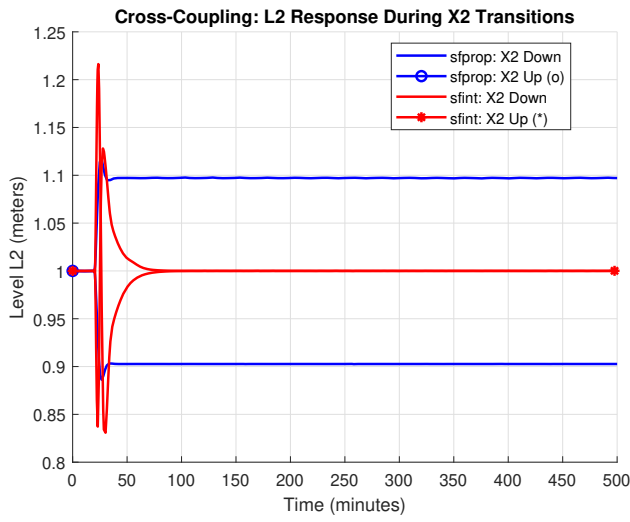
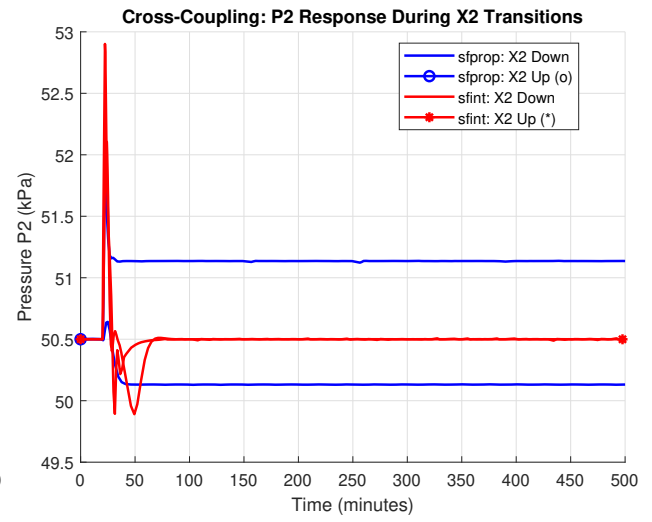


Figure 21: Individual Set point Changes



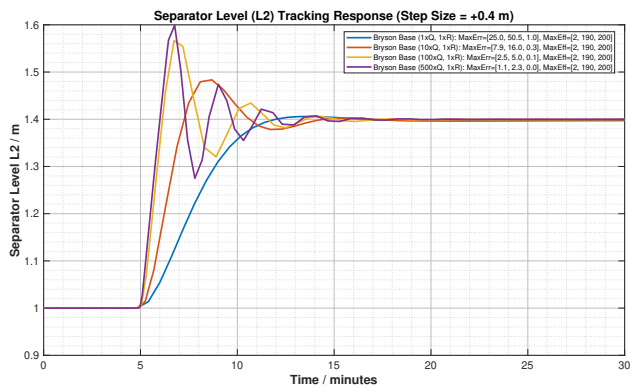
(a) L2 response to X2 set point change



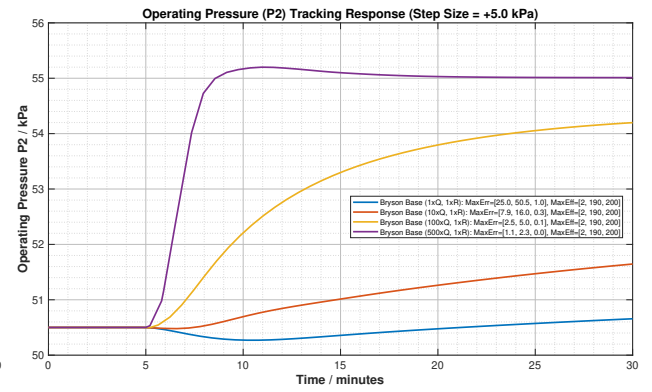
(b) P2 response to X2 set point change

Figure 22: Coupling Effect Comparison

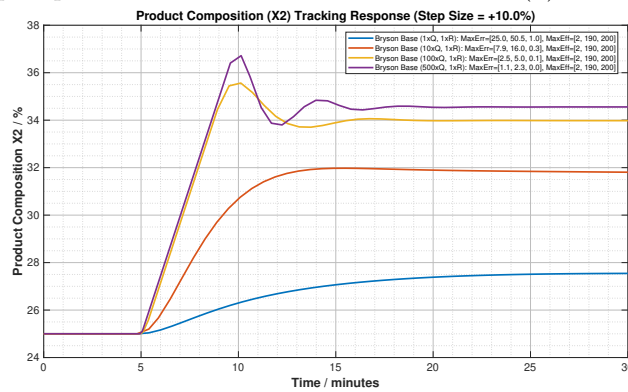
C.3 LQR



(a) L2 step response



(b) P2 step response



(c) X2 step response

Figure 23: LQR Tuning

D Variables and their nominal values

Table 3: Process variables and their nominal values

Variable	Description	Value & Units
F_1	Feed flowrate	10.0 kg/min
F_2	Product flowrate	2.0 kg/min
F_3	Circulating flowrate	50.0 kg/min
F_4	Vapour flowrate	8.0 kg/min
F_5	Condensate flowrate	8.0 kg/min
X_1	Feed composition	5.0 %
X_2	Product composition	25.0 %
T_2	Feed temperature	40.0°C
	Product temperature	84.6°C
T_3	Vapour temperature	80.6°C
L_2	Separator level	1.0 m
P_2	Operating pressure	50.5 kPa
F_{100}	Steam flowrate	9.3 kg/min
T_{100}	Steam temperature	119.9°C
P_{100}	Steam pressure	194.7 kPa
Q_{100}	Heater duty	339.0 kW
F_{200}	Cooling water flowrate	208.0 kg/min
T_{200}	Cooling water inlet temp	25.0°C
T_{201}	Cooling water outlet temp	46.1°C
Q_{200}	Condenser duty	307.9 kW